

GeoODE-KD: Distilling Sentence Embeddings as Teacher-Guided Geometric Dynamics

Abstract

Knowledge distillation provides an effective route for transferring the semantic capabilities of large text embedding models into compact encoders. Existing embedding distillation methods predominantly formulate knowledge transfer as *state matching*: the student is encouraged to reproduce teacher embeddings, intermediate representations, or pairwise relations at one or more discrete network depths. Such objectives specify where student representations should be located, but provide little structure on *how* they should evolve through depth. This limitation becomes particularly relevant under large teacher-student capacity gaps, where forcing shallow student layers toward highly abstract teacher representations can over-constrain their native feature organization.

We propose **GeoODE-KD**, a continuous-depth formulation of sentence embedding distillation that transfers *representation dynamics* rather than only representation states. We view normalized sentence embeddings as points on a hypersphere and construct a teacher-conditioned semantic energy that combines instance-level alignment with relational embedding geometry. This energy induces a Riemannian gradient flow whose vector field specifies a principled direction of semantic evolution toward the teacher space. Instead of introducing a Neural ODE module or an external numerical solver, we interpret the existing Transformer layers of the student as discrete integration steps of this continuous flow and regularize consecutive layers to follow the corresponding teacher-guided dynamics.

The resulting framework requires only cached final teacher embeddings, does not access teacher hidden states during student training, and introduces no additional module at inference time. We further establish that the continuous dynamics preserve embedding norms and monotonically decrease the teacher-conditioned distillation energy; in the instance-only case, cosine similarity to the teacher increases monotonically along continuous depth. We will evaluate whether this dynamics-based formulation improves teacher-student transfer, representation geometry, and downstream generalization over endpoint, multi-layer, relational, and flow-based distillation baselines.

1 Introduction

Dense text embeddings have become a fundamental interface between language models and downstream applications, supporting semantic search, retrieval-augmented generation, clustering, classification, and semantic textual similarity. Recent embedding models obtain strong general-purpose representations by increasing model size, training data, and representation dimensionality. However, these improvements come with substantial inference and deployment costs, motivating the compression of large embedding models into smaller encoders through knowledge distillation (Hinton et al., 2015; Reimers and Gurevych, 2020; Xu et al., 2023; Zhang et al., 2024; Truong et al., 2025).

A common principle underlying embedding distillation is *representation matching*. Given a teacher embedding τ and a student representation $\mathbf{z}$, the student is optimized to minimize a discrepancy such as cosine distance or mean squared error. More structured approaches enrich

this supervision using contrastive signals, intermediate hidden representations, attention relations, optimal transport, or pairwise geometry (Xu et al., 2023; Truong et al., 2025). Recent work further explores depth-wise supervision. For example, ESE encourages salient semantic information to emerge in shallower representations to support depth-scalable sentence embeddings (Li et al., 2025), while TALAS anchors selected upper student layers to the teacher output and propagates relational structure toward lower layers through adjacent-layer self-distillation (Anonymous).

Despite their differences, these approaches predominantly supervise a collection of *states*:

$$\mathbf{z}^{(1)}, \mathbf{z}^{(2)}, \dots, \mathbf{z}^{(L)}. \quad (1)$$

The corresponding objectives describe what one or more intermediate representations should resemble, but do not explicitly model the *transition* from one level of abstraction to the next. We argue that this distinction matters for embedding distillation. A shallow Transformer layer primarily captures local and lower-level linguistic structure, whereas deeper layers gradually accumulate task-relevant semantic information. Directly forcing many intermediate layers toward a fixed high-level teacher representation therefore risks destroying the student’s natural hierarchical organization. Indeed, recent layer-wise embedding distillation results indicate that stronger supervision at an increasing number of student layers does not necessarily produce better transfer, particularly under substantial teacher-student capacity mismatch (Anonymous).

This observation motivates a different question:

Instead of specifying where every student layer should be, can the teacher specify how the student’s representation should move through depth?

The continuous-depth perspective of residual networks provides a natural language for this question. Residual transformations can be interpreted as discrete numerical approximations to an ordinary differential equation (ODE), and this connection has motivated continuous-depth neural networks and ODE-inspired Transformer architectures (Chen et al., 2018; Li et al., 2022). Under this view, network depth plays the role of time and consecutive hidden states form a discrete trajectory,

$$\mathbf{z}^{(1)} \rightarrow \mathbf{z}^{(2)} \rightarrow \dots \rightarrow \mathbf{z}^{(L)}. \quad (2)$$

However, simply replacing Transformer layers with a generic Neural ODE does not by itself solve the embedding distillation problem. Moreover, continuous transformations have already been explored in knowledge transfer through diffusion and flow-matching mechanisms (Huang et al., 2023; Shao et al., 2024). Our goal is therefore different: we do not learn an auxiliary continuous network that transports a student output into a teacher output. Instead, we use the teacher to define a *semantic vector field*, and train the student’s existing discrete layers to realize that field.

We introduce **GeoODE-KD**, a teacher-guided geometric dynamics framework for sentence embedding distillation. Our starting point is that sentence embeddings are typically compared by cosine similarity and are commonly ℓ_2 -normalized. We therefore model student and teacher representations on the unit hypersphere $\mathbb{S}^{d-1}$ rather than in unconstrained Euclidean space. Given cached teacher embeddings, we construct a semantic potential energy consisting of two complementary terms. The first attracts each student sample toward its corresponding teacher representation, while the second matches the pairwise geometry of the student batch to that of the teacher. The negative Riemannian gradient of this energy defines a continuous vector field,

$$\frac{d\mathbf{Z}(t)}{dt} = -s(t) \operatorname{grad}_{\mathbb{S}} \mathcal{E}(\mathbf{Z}(t), \mathbf{T}), \quad (3)$$

which describes how sentence representations should evolve toward the teacher’s semantic space.

Crucially, GeoODE-KD does *not* numerically integrate this equation using an external ODE solver. We assign each Transformer layer a normalized depth coordinate t_ℓ and treat consecutive student representations as discrete observations of the continuous trajectory. At each depth, one Riemannian Euler step computed from the teacher-conditioned vector field provides the desired next representation. An ODE consistency objective then aligns the actual next Transformer layer with this predicted state. Consequently, the student itself becomes the numerical realization of the teacher-guided semantic flow.

This construction has several desirable properties. First, the teacher is used only to pre-compute final sentence embeddings; teacher inference and teacher hidden states are unnecessary during student optimization. Second, teacher supervision specifies a direction of semantic improvement rather than forcing every intermediate layer to imitate the same terminal representation. Third, because the dynamics live in the tangent space of the hypersphere, they respect the geometry induced by normalized sentence embeddings. Finally, all geometric and ODE-specific machinery is used only as a training objective; deployment requires exactly the original student encoder.

Our contributions are summarized as follows:

- We formulate sentence embedding distillation as *continuous representation dynamics*, shifting the target of knowledge transfer from isolated intermediate states to the evolution of representations through network depth.
- We introduce a teacher-conditioned Riemannian gradient flow on the embedding hypersphere. Its energy jointly captures point-wise semantic alignment and teacher relational geometry, yielding an explicit and interpretable semantic vector field without a learnable Neural ODE module.
- We propose an ODE consistency objective that treats the Transformer’s existing layers as discrete integration steps of the teacher-guided flow. The method requires only cached teacher output embeddings and incurs no additional inference-time architecture or ODE solver.
- We provide basic theoretical guarantees for the continuous dynamics, including hyperspherical norm preservation and monotonic distillation-energy descent. In the point-wise special case, we further show monotonic improvement in student-teacher cosine similarity along continuous depth.

2 Related Work

2.1 Knowledge Distillation for Sentence Embeddings

Knowledge distillation transfers predictive or representational knowledge from a high-capacity teacher to a compact student (Hinton et al., 2015). In sentence representation learning, Reimers and Gurevych (2020) demonstrated that sentence-level teacher representations can directly supervise a student embedding space, illustrating the effectiveness and simplicity of output-level embedding distillation. Such methods are particularly attractive when the teacher is large or accessible only through inference, since teacher embeddings can be computed offline.

Subsequent approaches enrich the distillation signal beyond direct point-wise imitation. DistillCSE incorporates knowledge distillation into contrastive sentence representation learning and studies the instability caused by high-variance teacher signals in contrastive settings (Xu et al., 2023). Jasper and Stella employ a multi-stage distillation strategy for transferring strong embedding models

into more deployable representations (Zhang et al., 2024). More recently, EMO addresses cross-tokenizer embedding distillation through intra-model relational alignment and optimal transport over hidden representations (Truong et al., 2025).

Our work retains the efficiency advantage of sentence-level teacher supervision but differs in what is distilled. Rather than using the final teacher embedding solely as a static target, GeoODE-KD uses it to define a vector field that supervises how student representations evolve through depth.

2.2 Layer-wise and Geometric Representation Transfer

Intermediate supervision has long been used to expose students to richer teacher information than final-output matching alone. In embedding models, this idea is complicated by architectural and capacity differences: representations at different depths need not encode semantic information at comparable abstraction levels.

ESE addresses depth efficiency by explicitly encouraging salient information to be represented at shallower layers, enabling sentence embedding models to operate at different depths (Li et al., 2025). TALAS takes a complementary distillation perspective: selected upper student layers are aligned to a cached final teacher embedding, while lower-layer knowledge propagation is encouraged by matching pairwise geometric relations between adjacent student layers (Anonymous). Its formulation highlights an important tension: direct teacher alignment can provide high-level semantics, but excessive anchoring may over-constrain lower or intermediate student representations.

GeoODE-KD builds on the broader insight that embedding geometry contains knowledge not captured by point-wise targets. However, rather than imposing a sequence of static layer constraints or requiring each adjacent layer to preserve the same relation matrix, we define a *global potential* conditioned on the teacher geometry. Its gradient determines the desired local transition at every student depth. Thus, relational knowledge affects not only the location of embeddings but also their direction of evolution.

2.3 Continuous-Depth Networks and ODE-Based Knowledge Transfer

Neural ODEs interpret hidden representations as states of a continuous dynamical system,

$$\frac{d\mathbf{h}(t)}{dt} = f_{\theta}(\mathbf{h}(t), t), \quad (4)$$

whose solution replaces an explicitly specified sequence of discrete layers (Chen et al., 2018). The connection between residual networks and numerical ODE integration has also been explored for Transformers. ODE Transformer, for example, relates Transformer residual computation to numerical integration and derives architectures inspired by higher-order ODE solvers (Li et al., 2022).

Continuous transformations have separately appeared in knowledge distillation. DiffKD models the student feature as a noisy version of the teacher feature and uses diffusion-based denoising before feature matching (Huang et al., 2023). Flow Matching provides a simulation-free framework for learning continuous normalizing flows (Lipman et al., 2023), and FM-KT applies continuous flow-based transport to progressively bridge teacher-student feature or logit distributions (Shao et al., 2024).

GeoODE-KD is conceptually distinct from both continuous-depth architecture design and learned flow-based feature transport. We neither replace the student architecture with a Neural ODE block nor train an auxiliary continuous normalizing flow that maps a student output to a teacher output. The student Transformer remains unchanged. Continuous dynamics instead serve as a *training*

principle: the teacher induces an analytic geometric vector field, and ordinary student layers are optimized to approximate its discretized trajectory. As a result, no ODE solver or flow module is required at inference time.

3 GeoODE-KD

3.1 Problem Formulation

Let

$$f_T : \mathcal{X} \rightarrow \mathbb{R}^{d_T}, \quad f_S : \mathcal{X} \rightarrow \mathbb{R}^{d_S} \quad (5)$$

denote a teacher embedding model and a compact student encoder, respectively. Given an unlabeled distillation corpus

$$\mathcal{D} = \{x_i\}_{i=1}^N, \quad (6)$$

the objective is to transfer the semantic structure of f_T into f_S without requiring task labels.

The student contains L Transformer layers. For sentence x_i , let $\mathbf{h}_i^{(\ell)}$ denote its hidden representation at layer ℓ . A pooling operator $\text{Pool}(\cdot)$ produces an intermediate sentence representation,

$$\mathbf{q}_i^{(\ell)} = \text{Pool}(\mathbf{h}_i^{(\ell)}). \quad (7)$$

We focus on the common large-teacher-small-student regime where $d_T \geq d_S$. Teacher outputs are transformed once into the student dimension using a fixed linear dimensionality reduction map $P_T \in \mathbb{R}^{d_T \times d_S}$. Our default choice is PCA fitted only on the cached training embeddings:

$$\boldsymbol{\tau}_i = \text{norm}(f_T(x_i)P_T), \quad (8)$$

where

$$\text{norm}(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|_2 + \epsilon}. \quad (9)$$

When teacher and student dimensions are equal, P_T is the identity.

Student representations are similarly normalized,

$$\mathbf{z}_i^{(\ell)} = \text{norm}(\mathbf{q}_i^{(\ell)}). \quad (10)$$

Therefore,

$$\mathbf{z}_i^{(\ell)}, \boldsymbol{\tau}_i \in \mathbb{S}^{d_S-1}, \quad (11)$$

where $\mathbb{S}^{d_S-1}$ denotes the unit hypersphere.

For a mini-batch of size B , we stack row-wise representations as

$$\mathbf{Z}^{(\ell)} = \begin{bmatrix} (\mathbf{z}_1^{(\ell)})^\top \\ \vdots \\ (\mathbf{z}_B^{(\ell)})^\top \end{bmatrix} \in \mathbb{R}^{B \times d_S}, \quad \mathbf{T} = \begin{bmatrix} \boldsymbol{\tau}_1^\top \\ \vdots \\ \boldsymbol{\tau}_B^\top \end{bmatrix}. \quad (12)$$

3.2 From Discrete Layer Matching to Continuous Semantic Dynamics

Conventional multi-layer distillation can be written schematically as

$$\mathcal{L}_{\text{layer}} = \sum_{\ell \in \mathcal{A}} D(\mathbf{Z}^{(\ell)}, \mathbf{T}), \quad (13)$$

where $\mathcal{A}$ denotes a selected set of student layers. Equation (13) treats each supervised layer as an independent state-matching problem.

We instead introduce a normalized depth coordinate

$$t_\ell = \frac{\ell}{L}, \quad \ell \in \{1, \dots, L\}, \quad (14)$$

and interpret

$$\mathbf{Z}^{(\ell)} \approx \mathbf{Z}(t_\ell) \quad (15)$$

as samples from an underlying continuous semantic trajectory.

Our objective is to construct a teacher-conditioned differential equation

$$\frac{d\mathbf{Z}(t)}{dt} = F(\mathbf{Z}(t), \mathbf{T}, t) \quad (16)$$

and train the discrete Transformer layers to approximate this flow.

3.3 Teacher-Conditioned Semantic Energy

The vector field in Eq. (16) should satisfy two properties. First, each sentence representation should progressively move toward the semantic target supplied by its teacher embedding. Second, the student should recover the relational geometry of the teacher embedding space rather than merely matching samples independently.

We encode these requirements through a teacher-conditioned energy function.

Instance-level semantic energy. We first define

$$\mathcal{E}_{\text{sem}}(\mathbf{Z}, \mathbf{T}) = \frac{1}{B} \sum_{i=1}^B (1 - \mathbf{z}_i^\top \boldsymbol{\tau}_i). \quad (17)$$

Since the representations are normalized, Eq. (17) is the average cosine distance between student and teacher embeddings.

Relational geometric energy. Sentence embeddings are primarily useful through relative similarities. We therefore construct batch-wise Gram matrices

$$\mathbf{G}_S = \mathbf{Z}\mathbf{Z}^\top, \quad \mathbf{G}_T = \mathbf{T}\mathbf{T}^\top. \quad (18)$$

Their (i, j) -th entries correspond to pairwise cosine similarities.

The geometric energy is

$$\mathcal{E}_{\text{geo}}(\mathbf{Z}, \mathbf{T}) = \frac{1}{B^2} \left\| \mathbf{Z}\mathbf{Z}^\top - \mathbf{T}\mathbf{T}^\top \right\|_F^2. \quad (19)$$

The total teacher-conditioned potential is

$$\boxed{\mathcal{E}(\mathbf{Z}, \mathbf{T}) = \alpha \mathcal{E}_{\text{sem}}(\mathbf{Z}, \mathbf{T}) + \beta \mathcal{E}_{\text{geo}}(\mathbf{Z}, \mathbf{T})} \quad (20)$$

with $\alpha, \beta \geq 0$.

Importantly, Eq. (20) is not directly minimized at every student layer. Instead, it defines the continuous dynamics that student layers are trained to realize.

3.4 Teacher-Guided Riemannian Gradient Flow

Because the embeddings lie on $\mathbb{S}^{d_S-1}$, unconstrained Euclidean motion is undesirable: an arbitrary update can alter vector norms and introduce radial components that are irrelevant to cosine geometry. We therefore construct a Riemannian gradient flow on the product of hyperspheres.

For an embedding $\mathbf{z}$, the tangent space is

$$T_{\mathbf{z}}\mathbb{S}^{d_S-1} = \{\mathbf{v} \in \mathbb{R}^{d_S} : \mathbf{z}^\top \mathbf{v} = 0\}. \quad (21)$$

The orthogonal projection of a Euclidean vector $\mathbf{u}$ onto this tangent space is

$$\Pi_{\mathbf{z}}(\mathbf{u}) = \mathbf{u} - (\mathbf{z}^\top \mathbf{u})\mathbf{z}. \quad (22)$$

For the batch representation, $\Pi_{\mathbf{Z}}$ applies Eq. (22) row-wise.

The Euclidean negative-gradient direction of the semantic energy is

$$-\nabla_{\mathbf{Z}}\mathcal{E}_{\text{sem}} = \frac{1}{B}\mathbf{T}, \quad (23)$$

whereas the geometric term satisfies

$$\nabla_{\mathbf{Z}}\mathcal{E}_{\text{geo}} = \frac{4}{B^2} \left(\mathbf{Z}\mathbf{Z}^\top - \mathbf{T}\mathbf{T}^\top \right) \mathbf{Z}. \quad (24)$$

We therefore define

$$\mathbf{U}(\mathbf{Z}, \mathbf{T}) = \frac{\alpha}{B}\mathbf{T} - \frac{4\beta}{B^2} \left(\mathbf{Z}\mathbf{Z}^\top - \mathbf{T}\mathbf{T}^\top \right) \mathbf{Z}, \quad (25)$$

$$F(\mathbf{Z}, \mathbf{T}, t) = s(t)\Pi_{\mathbf{Z}}[\mathbf{U}(\mathbf{Z}, \mathbf{T})]. \quad (26)$$

The continuous teacher-guided dynamics are then

$$\boxed{\frac{d\mathbf{Z}(t)}{dt} = F(\mathbf{Z}(t), \mathbf{T}, t) = -s(t) \text{grad}_{\mathbb{S}} \mathcal{E}(\mathbf{Z}(t), \mathbf{T})} \quad (27)$$

where $s(t) \geq 0$ controls the strength of teacher guidance with depth.

Depth-dependent semantic guidance. We use

$$s(t) = t \quad (28)$$

as the default schedule. This choice introduces weak teacher guidance at shallow depths and progressively strengthens it toward the output layers. The design reflects the intuition that shallow representations should retain flexibility to extract lower-level linguistic information, while deeper layers should become increasingly organized by the teacher’s semantic geometry.

More general schedules, e.g., $s(t) = t^p$ or a constant $s(t) = 1$, can be considered as ablations, but the linear schedule introduces no additional learned parameters.

3.5 Student Layers as Discrete ODE Steps

A central design choice of GeoODE-KD is that Eq. (27) is *not* solved using a black-box ODE solver. Instead, the student Transformer itself provides the discrete trajectory.

For consecutive student depths,

$$\Delta t = t_{\ell+1} - t_\ell = \frac{1}{L}. \quad (29)$$

Given $\mathbf{Z}^{(\ell)}$, one explicit Riemannian Euler step of the teacher-guided dynamics is

$$\tilde{\mathbf{Z}}^{(\ell+1)} = \text{Retr}_{\mathbf{Z}^{(\ell)}} \left(\Delta t F(\mathbf{Z}^{(\ell)}, \mathbf{T}, t_\ell) \right), \quad (30)$$

where we implement an efficient retraction using row-wise normalization:

$$\text{Retr}_{\mathbf{Z}}(\mathbf{V}) = \text{RowNorm}(\mathbf{Z} + \mathbf{V}). \quad (31)$$

The actual Transformer produces $\mathbf{Z}^{(\ell+1)}$. We regularize it toward the ODE-predicted next state:

$$\mathcal{L}_{\text{dyn}} = \frac{1}{L-1} \sum_{\ell=1}^{L-1} D_{\cos} \left(\mathbf{Z}^{(\ell+1)}, \text{sg} \left[\tilde{\mathbf{Z}}^{(\ell+1)} \right] \right), \quad (32)$$

where $\text{sg}[\cdot]$ denotes stop-gradient and

$$D_{\cos}(\mathbf{A}, \mathbf{B}) = \frac{1}{B} \sum_{i=1}^B \left(1 - \mathbf{a}_i^\top \mathbf{b}_i \right). \quad (33)$$

The stop-gradient operation makes $\tilde{\mathbf{Z}}^{(\ell+1)}$ a local dynamics target computed from the current trajectory, preventing the model from trivially modifying both sides of the consistency objective simultaneously. We will separately ablate full-gradient dynamics.

Equation (32) creates the main distinction between GeoODE-KD and static multi-layer distillation:

$$\text{State matching:} \quad \mathbf{Z}^{(\ell)} \rightarrow \mathbf{T}, \quad (34)$$

$$\text{GeoODE-KD:} \quad \mathbf{Z}^{(\ell)} \rightarrow \mathbf{Z}^{(\ell+1)} \quad \text{according to } F(\mathbf{Z}^{(\ell)}, \mathbf{T}, t_\ell). \quad (35)$$

The teacher thus supervises the student’s *velocity through depth* instead of demanding that every intermediate representation imitate the same terminal target.

3.6 Endpoint Semantic Distillation

While $\mathcal{L}_{\text{dyn}}$ determines the local trajectory, the student output should ultimately remain close to the teacher endpoint. We therefore apply direct distillation only at the final student layer:

$$\boxed{\mathcal{L}_{\text{end}} = \frac{1}{B} \sum_{i=1}^B \left(1 - (\mathbf{z}_i^{(L)})^\top \boldsymbol{\tau}_i \right)}. \quad (36)$$

Unlike conventional multi-layer teacher anchoring, no intermediate student layer is directly forced to equal the final teacher embedding. Intermediate supervision occurs only through the vector field in Eq. (26).

3.7 Contrastive Representation Regularization

Pure teacher imitation may transfer undesirable concentration or anisotropy from the teacher embedding distribution. We therefore retain a standard unsupervised contrastive objective at the final layer.

For each input x_i , two independent dropout masks produce normalized representations $\mathbf{z}_i^{(L,a)}$ and $\mathbf{z}_i^{(L,b)}$. We use

$$\mathcal{L}_{\text{ctr}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(\mathbf{z}_i^{(L,a)}, \mathbf{z}_i^{(L,b)})/\tau_c)}{\sum_{j=1}^B \exp(\text{sim}(\mathbf{z}_i^{(L,a)}, \mathbf{z}_j^{(L,b)})/\tau_c)}, \quad (37)$$

where τ_c is the contrastive temperature.

Importantly, this objective is applied only to the final representation. We avoid imposing contrastive targets independently at every layer, which would confound the role of continuous dynamics.

3.8 Overall Training Objective

The final GeoODE-KD objective is

$$\boxed{\mathcal{L}_{\text{total}} = \lambda_{\text{end}} \mathcal{L}_{\text{end}} + \lambda_{\text{dyn}} \mathcal{L}_{\text{dyn}} + \lambda_{\text{ctr}} \mathcal{L}_{\text{ctr}}}. \quad (38)$$

The method introduces two conceptually distinct hyperparameters inside the semantic potential: α controls point-wise teacher attraction and β controls relational geometric preservation. For the initial implementation, we set $\alpha = 1$ and tune only β , while fixing $\lambda_{\text{end}} = \lambda_{\text{dyn}} = 1$ and tuning λ_{ctr} on a validation split.

We emphasize the separation between Eq. (20) and Eq. (38). The energy $\mathcal{E}$ is not simply another layer-wise training loss. Instead, it defines the vector field that specifies the desired *direction of change*, whereas $\mathcal{L}_{\text{dyn}}$ measures whether the Transformer’s actual layer transition follows that direction.

3.9 Theoretical Properties

We next state basic properties that motivate the proposed dynamics.

Proposition 1: Hyperspherical norm preservation. Consider a single trajectory $\mathbf{z}(t) \in \mathbb{S}^{d-1}$ governed by

$$\dot{\mathbf{z}} = \Pi_{\mathbf{z}}(\mathbf{u}). \quad (39)$$

Then

$$\frac{d}{dt} \|\mathbf{z}(t)\|_2^2 = 0. \quad (40)$$

Proof. By construction, $\Pi_{\mathbf{z}}(\mathbf{u})$ lies in the tangent space of the hypersphere and therefore

$$\mathbf{z}^\top \Pi_{\mathbf{z}}(\mathbf{u}) = 0. \quad (41)$$

Consequently,

$$\frac{d}{dt} \|\mathbf{z}\|_2^2 = 2\mathbf{z}^\top \dot{\mathbf{z}} = 0. \quad \square \quad (42)$$

Thus, the continuous teacher-guided dynamics alter semantic direction without introducing an irrelevant radial component.

Proposition 2: Monotonic teacher-conditioned energy descent. Let the continuous trajectory follow

$$\dot{\mathbf{Z}} = -s(t) \text{grad}_{\mathbb{S}} \mathcal{E}(\mathbf{Z}, \mathbf{T}), \quad (43)$$

where $s(t) \geq 0$. Then

$$\frac{d}{dt} \mathcal{E}(\mathbf{Z}(t), \mathbf{T}) \leq 0. \quad (44)$$

Proof. Using the Riemannian chain rule,

$$\frac{d\mathcal{E}}{dt} = \left\langle \text{grad}_{\mathbb{S}} \mathcal{E}, \dot{\mathbf{Z}} \right\rangle \quad (45)$$

$$= -s(t) \|\text{grad}_{\mathbb{S}} \mathcal{E}\|_F^2 \leq 0. \quad \square \quad (46)$$

Hence, the ideal continuous trajectory monotonically reduces the combined semantic and relational discrepancy from the teacher.

Corollary 1: Monotonic teacher similarity under point-wise flow. Consider $\beta = 0$ and a single normalized teacher-student pair $(\boldsymbol{\tau}, \mathbf{z})$. Ignoring the positive constant α/B , the dynamics reduce to

$$\dot{\mathbf{z}} = s(t) \left[\boldsymbol{\tau} - (\mathbf{z}^\top \boldsymbol{\tau}) \mathbf{z} \right]. \quad (47)$$

Define

$$c(t) = (\mathbf{z}(t))^\top \boldsymbol{\tau}. \quad (48)$$

Then

$$\dot{c}(t) = (\dot{\mathbf{z}}(t))^\top \boldsymbol{\tau} \quad (49)$$

$$= s(t) \left[1 - (c(t))^2 \right] \geq 0. \quad (50)$$

Therefore, unless the student is already parallel or antiparallel to the teacher, the continuous point-wise dynamics monotonically increase student-teacher cosine similarity.

Equation (50) provides an interpretable notion of semantic progression: deeper continuous states are guaranteed to become increasingly aligned with the teacher under the instance-only field. When relational geometry is included, individual cosine similarity need not be monotonic, but Proposition 2 guarantees monotonic descent of the joint semantic-geometric energy.

3.10 Training Algorithm

3.11 Training and Inference Complexity

Teacher inference is entirely offline. For a corpus of N examples, only one final teacher embedding per training example is stored. During student training, GeoODE-KD does not require teacher hidden states, attention maps, or parallel teacher forward passes.

For a batch of B samples and embedding dimension d_S , computing the relational component at one layer requires $\mathcal{O}(B^2 d_S)$ operations for the Gram matrix and $\mathcal{O}(B^2)$ temporary relation memory. Across L student layers, the dynamics objective therefore adds approximately

$$\mathcal{O}(LB^2 d_S) \quad (51)$$

training computation. This cost is independent of the size of the teacher.

Algorithm 1 GeoODE-KD Training

Require: Unlabeled corpus $\mathcal{D}$, teacher f_T , student f_S with L layers, $\alpha, \beta, \lambda_{\text{end}}, \lambda_{\text{dyn}}, \lambda_{\text{ctr}}$

- 1: Run f_T once over $\mathcal{D}$ and cache teacher embeddings
 - 2: Fit fixed PCA map P_T on cached teacher embeddings
 - 3: Transform and normalize all cached targets using Eq. (8)
 - 4: **for** each training mini-batch $\mathcal{B}$ **do**
 - 5: Retrieve cached teacher matrix $\mathbf{T}$
 - 6: Run the student and obtain $\{\mathbf{Z}^{(\ell)}\}_{\ell=1}^L$
 - 7: $\mathcal{L}_{\text{dyn}} \leftarrow 0$
 - 8: **for** $\ell = 1, \dots, L - 1$ **do**
 - 9: $t_\ell \leftarrow \ell/L$
 - 10: Compute $\mathbf{G}_S \leftarrow \mathbf{Z}^{(\ell)}(\mathbf{Z}^{(\ell)})^\top$
 - 11: Compute $\mathbf{G}_T \leftarrow \mathbf{T}\mathbf{T}^\top$
 - 12: Compute vector field $F(\mathbf{Z}^{(\ell)}, \mathbf{T}, t_\ell)$ using Eq. (26)
 - 13: Predict $\tilde{\mathbf{Z}}^{(\ell+1)}$ using Eq. (30)
 - 14: Accumulate ODE consistency using Eq. (32)
 - 15: **end for**
 - 16: Compute endpoint loss $\mathcal{L}_{\text{end}}$ using Eq. (36)
 - 17: Obtain a second dropout view and compute $\mathcal{L}_{\text{ctr}}$
 - 18: Compute $\mathcal{L}_{\text{total}}$ using Eq. (38)
 - 19: Update student parameters with AdamW
 - 20: **end for**
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At inference time, all teacher-conditioned components are removed:

$$x \xrightarrow{f_S} \mathbf{z}^{(L)}. \quad (52)$$

No PCA target projection, Gram matrix, semantic energy, vector field, ODE solver, or auxiliary network is evaluated. GeoODE-KD therefore has the same inference architecture and asymptotic inference cost as its student backbone.

3.12 Relationship to Alternative Distillation Objectives

GeoODE-KD can be understood by contrasting four forms of supervision.

Endpoint KD.

$$\mathbf{Z}^{(L)} \rightarrow \mathbf{T}. \quad (53)$$

Only the final state is constrained.

Static multi-layer KD.

$$\mathbf{Z}^{(\ell)} \rightarrow \mathbf{T} \quad \forall \ell \in \mathcal{A}. \quad (54)$$

Multiple states are constrained toward a common endpoint.

Adjacent relational self-distillation.

$$\mathbf{G}^{(\ell)} \rightarrow \mathbf{G}^{(\ell+1)}. \quad (55)$$

Local student geometries are encouraged to remain consistent across depth.

GeoODE-KD.

$$\mathbf{Z}^{(\ell+1)} \approx \text{Retr}_{\mathbf{Z}^{(\ell)}} \left[\Delta t F(\mathbf{Z}^{(\ell)}, \mathbf{T}, t_\ell) \right]. \quad (56)$$

Here, the teacher specifies a *directional transformation* at each depth. The student is not required to arrive at the final teacher representation prematurely; it is required to move in a direction that continuously decreases a teacher-conditioned semantic potential.

Research hypothesis. Our central hypothesis is that a compact student benefits more from learning a *semantic trajectory* than from repeatedly imitating a high-capacity teacher state. If correct, GeoODE-KD should exhibit three observable behaviors. First, student-teacher semantic discrepancy should decrease more smoothly through network depth than under static multi-layer distillation. Second, the teacher-student relational geometry gap should contract progressively across layers rather than only at the output. Third, these improvements should translate into better downstream embedding quality without requiring stronger inference-time computation. The experiments will directly test each of these hypotheses rather than evaluating downstream accuracy alone.

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